Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space. Because sensor telemetry contains unforecastable jitter and scale disparities, observation-space residuals develop heavy tails that trigger severe false-alarm flooding under automated threshold calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid reconstructing raw noise by forecasting compact latent summaries of future windows. We investigate how latent prediction conditions residual distributions under regime shift. Specifically, we compare a single-horizon baseline (TS-JEPA) against Soliton-Reynolds JEPA, a multi-horizon architecture that taps intermediate temporal convolutional stages, subtracts online context barycenters, whitens multi-horizon prediction errors, and scores centered sub-horizon fluctuations via a Lieb–Thirring spectral concentration ratio. Across continuous physical telemetry benchmarks calibrated on held-out nominal data with Streaming Peaks-Over-Threshold (SPOT), latent scoring suppresses false alarms under both retrospective buffered attribution and strictly causal streaming while improving anomaly ranking and event recall. Controlled factorial ablations isolate four distinct mechanisms: layer-normalization bottlenecks compress residual kurtosis, learned latent projections and multi-horizon barycenter whitening curb calibration-to-holdout drift, short prediction horizons eliminate emission lag, and hybrid sensor-variance monitors recover quiet single-channel faults.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable healthcare [1]–[3]. In NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels track power buses, thermal gradients, vibrations, and kinematics [4]. Monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: In deployed monitoring, an automated rule must convert continuous anomaly scores into binary alarms using healthy (nominal) calibration data alone. Standard deep detectors (TimesNet, TranAD, Anomaly Transformer) train a neural network to reconstruct every raw sensor reading and flag timestamps where the point-by-point reconstruction error is large [5]–[7]. Because raw telemetry contains random sensor jitter and scale differences across channels, those point-level errors produce frequent, clustered spikes, resulting in a heavy-tailed residual distribution with high excess kurtosis ($\gamma_2 > 8$, Table I). When paired with an automated thresholding rule such as Streaming Peaks-Over-Threshold (SPOT) [8] set to a 0.1% target false-alarm rate ($q = 10^{-3}$), normal background shifts in unseen data easily cross the threshold, triggering false alarms on 22%–31% of healthy timesteps.

Latent Residual Conditioning and Scope of Claims: Predicting a compact *summary* of a future time window, rather than reconstructing every noisy sensor sample, filters out random jitter before the prediction error is computed. We study Joint-Embedding Predictive Architectures (JEPAs) [9], which compress the preceding 256 timesteps into a 32-dimensional summary vector and forecast the summary of the next S timesteps ($S = 64$ by default, or $S \in \{1, 4\}$ for fast-response tripwires) using healthy training data alone [10]. Alongside a single-horizon baseline (TS-JEPA) that predicts the entire future window at once, we introduce *Soliton-Reynolds JEPA*, a multi-horizon predictor that monitors four nested lookahead windows simultaneously. To catch sharp fault onsets without being fooled by slow background drift, *Soliton-Reynolds JEPA* combines features from multiple depths of its temporal convolutional network [11], subtracts the current batch average (barycenter) to cancel slow level shifts, and standardizes (whitens) prediction errors across the four sub-horizons. Drawing on Reynolds decomposition [12] and Lieb–Thirring spectral bounds [13], it then separates those multi-horizon errors into an average shift and sub-horizon fluctuations, measuring whether fluctuation energy is spread evenly across horizons like background noise or concentrated into a coherent wavepacket like an emerging fault. Rather than claiming that static thresholds achieve an exact 0.1% false-alarm rate on non-stationary streams, we show that latent prediction reduces false alarms, curbs distribution drift between calibration and test data, and improves anomaly ranking. We evaluate four questions: (i) how much latent residuals lower realized false-positive rates (FPRs) and out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ under static, declustered, and rolling thresholds; (ii) how normalization, learned latent targets, multi-horizon whitening, and wavepacket concentration each contribute to

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

Color versions of one or more figures in this article are available online at https://ieeexplore.ieee.org.

those gains; (iii) how detection varies across point spikes, in-range contextual faults, and quiet single-sensor freezes; and (iv) how shortening the lookahead horizon S trades off alarm lag ($S-1$) against detection quality.

A. Contributions

1) *Why latent predictors stabilize alarm thresholds:* Across 12 matched-backbone models (272k–312k online parameters), we separate two distinct mechanisms. First, a layer-normalization (LayerNorm) [14] bottleneck caps extreme error spikes, compressing tail heaviness (excess kurtosis) from 6.10 to 0.91. Second, predicting a learned latent summary instead of raw sensor waveforms cuts distribution drift between calibration and later healthy data in half (D_{KS} drops from 0.376 to 0.214 in TS-JEPA), and multi-horizon barycenter whitening in Soliton-Reynolds JEPA lowers drift further to $D_{\mathrm{KS}} = 0.179$ (Section VI-D, Table II(c)).
2) *Fewer false alarms, contextual-anomaly dominance, and quiet-fault recovery:* Across the 19 non-GHL benchmark streams, latent scoring cuts held-out FPRs three-fold under retrospective buffered attribution (7.8%–9.3% vs. 30.0%–30.7% for TimesNet and TranAD) and by more than an order of magnitude under strictly causal streaming (0.6%–2.4% vs. 22.2%–24.3%). Soliton-Reynolds JEPA matches TS-JEPA on overall buffered precision-recall area under the curve (PR-AUC 0.331), catches 56.2% of all anomaly events at an equalized 1% false-alarm budget, and leads on NASA's ground-truth contextual anomalies (0.652 buffered PR-AUC, 75.0% causal SPOT recall at 0.6% FPR, and 82.0%–92.0% recall at 1%–5% FPR budgets). Pairing latent prediction with a rolling sensor-variance check further recovers quiet single-channel freezes (Tables II(c), III, and IV, Figs. 3(b) and 4(b)).
3) *Multi-horizon wavepacket bounds, ablation controls, and fast response:* Proposition 1 proves that after subtracting the average error across M sub-horizons, the Lieb–Thirring spectral concentration ratio stays strictly between $\frac{1}{M-1}$ (errors spread evenly across horizons like background noise) and 1 (errors aligned into a single fault wavepacket), regardless of signal amplitude. Controlled ablations on the same trained network show that multi-horizon whitened errors drive the ranking and drift improvements, whereas the post-hoc wavepacket multiplier has a negligible aggregate effect under Two One-Sided Tests (TOST) [15] equivalence. For time-critical monitoring, short horizons ($S \in \{1, 4\}$) eliminate the 63-step window lag while keeping false alarms low (Tables II and III, Fig. 4(a)).
4) *Why automated thresholds fail and when they hold:* By comparing chronological evaluation against block-shuffled and randomly shuffled healthy data with bootstrap confidence intervals (CIs), we separate slow background drift from short-term correlation. Slow regime drift between calibration and test periods causes 71.3% of the excess causal false alarms in single-horizon TS-JEPA and 78.3%–81.8% in observation-space reconstructors. Subtracting the batch barycenter in Soliton-Reynolds JEPA cuts this inter-block drift share four-fold to 18.7%, and runs-declustered SPOT or rolling quantiles further stabilize thresholds under step and variance shifts (Table V).

TABLE I
KEY OPERATIONAL AND STATISTICAL TERMS.

Term	Meaning
Latent residual	Gap between the predicted and observed 32-dimensional summary of the next window.
SPOT	Streaming extreme-value rule that sets an alarm threshold from healthy calibration scores alone.
Calibration error	Gap $\lvert \mathrm{FPR}_{\mathrm{hold}} - q \rvert$ between the actual false-alarm rate on healthy test data and the 0.1% target ($q{=}10^{-3}$).
Event recall	Fraction of true anomaly segments with at least one alarm.
Buffered	Retrospective scoring that averages each window score across its S steps.
Causal	Real-time scoring that uses only past data and assigns each score to its final step $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance.
Excess kurtosis	Tail heaviness of the residual distribution; large values mean frequent, extreme spikes.
Soliton ratio	Score $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ measuring whether sub-horizon errors align into one coherent wavepacket.

Roadmap. Section II reviews related work and Section III details the benchmark cohort. Section IV formulates the architectures, Section V defines the protocol, Sections VI and VII report ablations and benchmark results, and Section VIII concludes.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [7] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [5] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [6] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that destabilize extreme-value threshold fitting under baseline shift [16].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [17]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [18] and TS2Vec [19] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [20] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11

datasets, Table III(d)). Joint-Embedding Predictive Architectures (JEPAs) [9], [21], [22] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} produced by a separate target encoder. Collapse is prevented via exponential moving average (EMA) target updates [23] or variance-covariance penalties such as VICReg [24], Barlow Twins [25], and batch whitening [26]. How latent predictive residuals behave under unsupervised extreme-value threshold calibration remains unexplored.

Fluid-Dynamic Reynolds Decomposition and Spectral Concentration Priors: Multi-sensor telemetry reflects physical dynamical trajectories [27]. Neural ordinary differential equations (ODEs) [28], Physics-Informed Neural Networks (PINNs) [29], and Hamiltonian or Lagrangian networks [30], [31] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [32]. Koopman and operator-theoretic models [33]–[37] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we draw on Reynolds decomposition [12] from turbulence modeling to separate multi-horizon latent predictions into a mean trajectory and centered sub-horizon fluctuations. We then measure wavepacket localization across prediction horizons via a Lieb–Thirring spectral concentration ratio [13] on the centered fluctuation Gram matrix.

Extreme-Value Threshold Calibration and Evaluation Rigor: Grounded in the Pickands–Balkema–de Haan extreme-value theorem [38], Streaming Peaks-Over-Threshold (SPOT) [8] fits a Generalized Pareto Distribution (GPD) to score exceedances above quantile u to extrapolate a threshold at design risk q. However, continuous telemetry violates the classical independent and identically distributed (i.i.d.) assumption through short-range serial correlation (quantified by the Ferro–Segers extremal index [39]) and inter-block regime drift. Because "point-adjusted" F1 severely inflates scores [16], [40], [41], we evaluate all models using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm operating curves.

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [42], [43], TimeEval [16], and mTSBench [44]. Thirty-three streams enter the primary multi-seed held-out protocol; 12 are reserved with explicit accounting below. The suite spans four physical domains and separates natural operational or physiological anomalies from testbed or simulated faults. *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [45] (K=38), all containing natural operational or solar-flare events; on SMAP and MSL, NASA's ground-truth archive [4] further classifies each anomaly sequence as either an out-of-range *point* fault or an in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [42] sampled at 64 Hz (K=9, 72 natural gait-freezing episodes, where S=64 equals a 1 s horizon). *Industrial processes and smart infrastructure* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 testbed valve and heater faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 natural hydrological anomalies), Metro I-94 hourly highway traffic (K=5), Room-Occupancy sensing [46] (K=5), CalIt2 building flow [43] (K=2), and six CICIDS network streams (K=72, simulated attack flows). Cohort 3 (Section VI-B) separately injects six controlled single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [4], [16].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table III(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9, 1 held-out event) enters Cohorts 3–4. To eliminate subset-selection bias, Cohort 5 evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 Opportunity, GECCO, Genesis, Pooled Server Metrics (PSM), credit-card, Metro, and SWAN) at three seeds alongside the 130-stream baseline screen (Table III(d)).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [41]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Joint-Embedding Predictive Architectures and Soliton-Reynolds Decomposition

A. Problem Formulation and Latent Space Prediction

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to calibrated binary alarms. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and splits the series at index t into: (i) a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$), establishing the nominal regime; and (ii) a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t+S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$), inspected for trajectory

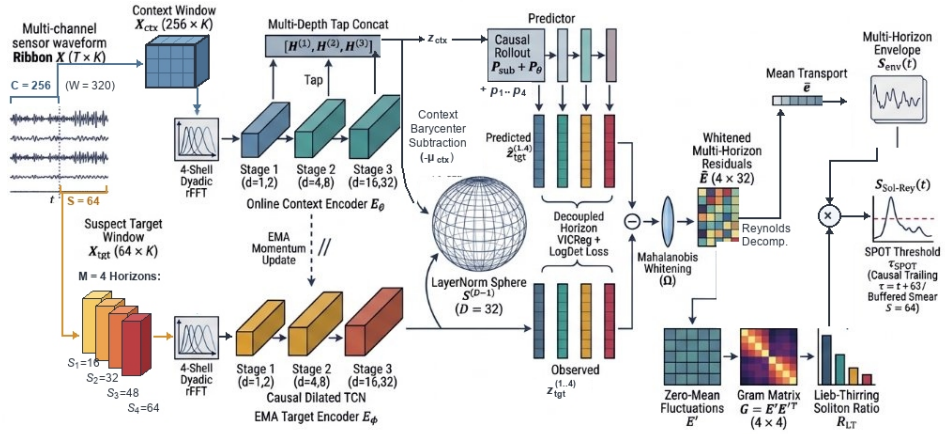


Fig. 1. Overview of the four-stage Soliton-Reynolds JEPA pipeline: (1) multiscale windowing and causal encoding, (2) dual-encoder hierarchy tapping and barycenter centering, (3) nested multi-horizon latent rollout, and (4) whitened residual scoring and extreme-value threshold calibration.

departures. Each window passes through a 4-shell dyadic real fast Fourier transform (rFFT) front-end and a causal dilated temporal convolutional network (TCN) [11].

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit leptokurtic sensor noise (Sections I and II), Stage 2 shifts prediction into a 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders extract three multi-depth taps across the shallow, mid, and deep TCN hierarchy and map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ into spherical representations $z_{\text{ctx}}, z_{\text{tgt}} \in \sqrt{D}\mathbb{S}^{D-1}$. To prevent constant-vector collapse, target weights ϕ are updated as an exponential moving average (EMA) of online weights θ, $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with cosine momentum $m_k \in [0.995, 0.9995]$ [21], [23]. Stage 3 then rolls out latent predictions across $M = 4$ nested sub-horizons ($S_m \in \{16, 32, 48, 64\}$) under a two-term multi-horizon mean-squared error (MSE) and covariance whitening defect objective (Section IV-B), and Stage 4 applies Mahalanobis residual whitening, Lieb–Thirring soliton concentration modulation, and coordinate-saliency-gated SPOT calibration under causal trailing ($\tau = t + 63$) and buffered smear mappings (Section IV-C).

B. Latent Predictor Architectures: TS-JEPA and Soliton-Reynolds JEPA

We evaluate two nested latent predictors sharing the same multiscale encoder backbone and coordinate saliency gate: an unconstrained single-horizon baseline (TS-JEPA) and a multi-horizon fluid-dynamic decomposition model (Soliton-Reynolds JEPA).

1) Unconstrained TS-JEPA Baseline: In TS-JEPA, a two-layer multi-layer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-horizon prediction $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with standard single-horizon predictive alignment and VICReg variance-covariance penalties [24] ($\gamma_{\text{var}} = 1.0$). Terminal LayerNorm [14] bounds latent representations on the sphere of radius $\sqrt{D}$ ($\|z\|_2 = \sqrt{D}$) and compresses nominal residual kurtosis relative to observation-space reconstructors. However, single-horizon encoding pools the entire $S = 64$ target window into one terminal TCN state, smoothing out short-lived sub-horizon transients and leaving cross-horizon residual correlations unwhitened.

2) Soliton-Reynolds JEPA (Multi-Horizon Reynolds Decomposition and Lieb–Thirring Concentration): Physical Motivation and Structural Design: In fluid turbulence, Reynolds decomposition [12] separates a velocity field into its time-averaged mean flow and zero-mean fluctuations, while dispersive wave theory characterizes localized, shape-preserving wavepackets (*solitons*) by spectral concentration of the fluctuation operator [13]. Soliton-Reynolds JEPA brings both principles into latent time-series prediction through two architectural stages before residual scoring:

1) *Multi-depth TCN functional hierarchy taps and synchronized barycenter centering:* Standard TCN encoders read only the deepest dilated block, where wide receptive fields attenuate sharp onset transients. In Soliton-Reynolds JEPA, a multi-depth tap head extracts $L_{\text{tap}} = 3$ intermediate feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) spanning the shallow dyadic front-end, mid-depth dilated block, and terminal TCN block. For each tap, the centered terminal state and temporal root-mean-square energy are concatenated into $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, mapped into $\mathbb{R}^D$ via a two-layer projection, and fused with the deep bottleneck representation via a bounded gate $g = 0.5\,\sigma(\theta_g) \le 0.5$. To eliminate common-mode baseline drift between context and target windows, the online running context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$ is subtracted from both online context and EMA target fused embeddings prior to final LayerNorm projection onto $\sqrt{D}\mathbb{S}^{D-1}$.
2) *Nested causal multi-horizon rollout and covariance whitening defect regularization:* Rather than predicting only the full $S = 64$ window, we partition $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefix windows $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ with lengths $S_m = (m/M)S \in \{16, 32, 48, 64\}$. The EMA target encoder encodes each prefix into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. From z_{ctx}, a sub-horizon predictor P_{sub} rolls out intermediate predictions $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \dots, M-1\}$ ($\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$) using learnable horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the primary predictor P_θ forecasts the terminal horizon $\hat{z}_{\text{tgt}}^{(M)} = \text{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ for $m < M$ and $w_M = \frac{1}{2}$, the model is trained with a parsimonious two-term objective combining multi-horizon predictive MSE

and the Frobenius covariance whitening defect:

$$\mathcal{L}_{\text{Sol-Rey}} = \sum_{m=1}^{M} \frac{w_m}{D} \Big(\big\| \hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big) \big\|_2^2 + \big\| \boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big) \big\|_F^2 \Big), \tag{1}$$

where $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ denotes the covariance whitening defect matrix of centered batch predictions $\mathbf{Z}_c = \mathbf{Z} - \mathbf{1}_B \bar{\mathbf{z}}^\top$ with sample covariance $\boldsymbol{\Sigma}(\mathbf{Z}) = (B-1)^{-1}\mathbf{Z}_c^\top \mathbf{Z}_c$ and coordinate standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + \epsilon}$ ($\gamma = 0.75$, $\epsilon = 10^{-4}$), defined by diagonal entries $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dd} = \sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ and off-diagonal entries $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dj} = \frac{1}{\sqrt{2}}\Sigma_{dj}(\mathbf{Z})$ for $d \neq j$. Unlike Cholesky log-determinant barriers that become rank-deficient when the minibatch size is smaller than the embedding dimension ($B = 16 < D = 32$), $\|\boldsymbol{\Delta}_\gamma(\mathbf{Z})\|_F^2$ remains smooth and full-rank along all D diagonal coordinates for any $B \geq 2$ while penalizing off-diagonal cross-correlations (achieving effective rank $\text{RankMe}(z_{\text{tgt}}) = 19.8$ [47] vs. 11.6 for single-horizon baselines).

Reynolds Fluctuation Gram Matrix and Lieb–Thirring Soliton Concentration: Physically, nominal background fluctuations disperse energy across multiple uncorrelated sub-horizons, whereas an emerging fault wavepacket concentrates fluctuation energy into a single dominant rank-1 directional mode across horizons. To formalize this distinction, let $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_M]^\top \in \mathbb{R}^{M\times D}$ stack the M prediction residuals $\mathbf{e}_m = z_{\text{tgt}}^{(m)} - \hat{z}_{\text{tgt}}^{(m)}$. We fit the empirical shrinkage precision matrix $\boldsymbol{\Omega} = (\tilde{\boldsymbol{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [48] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows to obtain Mahalanobis-whitened residuals $\tilde{\mathbf{e}} = \boldsymbol{\Omega}(\text{vec}(\mathbf{E}) - \bar{\boldsymbol{\mu}}_{\text{res}}) \in \mathbb{R}^{MD}$, reshaped into $\tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \ldots, \tilde{\mathbf{e}}_M]^\top \in \mathbb{R}^{M\times D}$. Applying Reynolds decomposition across the M horizons separates $\tilde{\mathbf{E}}$ into its horizon-mean transport $\bar{\mathbf{e}} = M^{-1}\sum_{m=1}^{M}\tilde{\mathbf{e}}_m \in \mathbb{R}^D$ and zero-mean fluctuations $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M\bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. From the $M \times M$ fluctuation Gram matrix $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ with ordered eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$, we define the Lieb–Thirring spectral concentration ratio [13] of order $\gamma = 1$:

$$R_{\text{LT}}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M}\lambda_m^{\gamma+1/2} + \epsilon_{\text{LT}}}, \quad \epsilon_{\text{LT}} = 10^{-12}. \tag{2}$$

Proposition 1 establishes the exact algebraic bounds and radial scale invariance of $R_{\text{LT}}(\mathbf{E}')$.

Proposition 1 (Lieb–Thirring Soliton Concentration Bounds, Centered Gram Rank Deficiency, and Scale Invariance)**.** *Let $\mathbf{E}' = (\mathbf{I}_M - M^{-1}\mathbf{1}_M\mathbf{1}_M^\top)\tilde{\mathbf{E}} \in \mathbb{R}^{M\times D}$ denote the Reynolds-centered multi-horizon fluctuation matrix for $M \geq 2$ and $\gamma > -1/2$. In the exact limit $\epsilon_{LT} \to 0^+$ for any non-zero fluctuation $\mathbf{E}' \neq \mathbf{0}$: (i)* Centered Gram nullspace and effective rank $M-1$: *Because $\mathbf{E}'^\top\mathbf{1}_M = \mathbf{0}$, the constant vector $\mathbf{1}_M$ lies in* $\ker(\mathbf{G})$*, forcing $\lambda_1(\mathbf{G}) \equiv 0$ identically so that* $\text{rank}(\mathbf{G}) \leq \min(M-1, D)$*. (ii)* Sharp isotropy lower bound and soliton upper bound: *The concentration ratio satisfies the tight two-sided inequality*

$$\frac{1}{M-1} \leq R_{LT}(\mathbf{E}') \leq 1, \tag{3}$$

where the lower bound $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) is attained if and only if fluctuation energy is distributed isotropically across $M-1$ orthogonal sub-horizon modes ($\lambda_2 = \cdots = \lambda_M > 0$), and the upper bound 1 is attained if and only if $\text{rank}(\mathbf{E}') = 1$ *($\lambda_2 = \cdots = \lambda_{M-1} = 0 < \lambda_M$), corresponding to a coherent rank-1 soliton wavepacket. (iii)* Exact radial scale invariance: *For any scalar $c > 0$, $R_{LT}(c\mathbf{E}') = R_{LT}(\mathbf{E}')$, rendering R_{LT} invariant to homogeneous amplitude rescaling.*

Proof. By construction, $\mathbf{G}\mathbf{1}_M = \mathbf{E}'(\mathbf{E}'^\top\mathbf{1}_M) = \mathbf{0}$, so $\lambda_1 = 0$ is always an eigenvalue of $\mathbf{G} \succeq \mathbf{0}$, leaving at most $M-1$ positive eigenvalues $\lambda_2 \leq \cdots \leq \lambda_M$ with $\lambda_M > 0$ since $\mathbf{E}' \neq \mathbf{0}$. Because $x \mapsto x^{\gamma+1/2}$ is strictly increasing on $x \geq 0$ for $\gamma > -1/2$, $\lambda_m^{\gamma+1/2} \leq \lambda_M^{\gamma+1/2}$ for all $m \in \{2, \ldots, M\}$. Summing over the $M-1$ remaining modes gives $\lambda_M^{\gamma+1/2} \leq \sum_{m=2}^{M}\lambda_m^{\gamma+1/2} \leq (M-1)\lambda_M^{\gamma+1/2}$. Dividing through by $\sum_{m=2}^{M}\lambda_m^{\gamma+1/2}$ yields (3), with equality on the left iff $\lambda_2 = \cdots = \lambda_M$ and on the right iff $\lambda_2 = \cdots = \lambda_{M-1} = 0$. Finally, scaling $\mathbf{E}' \mapsto c\mathbf{E}'$ maps $\mathbf{G} \mapsto c^2\mathbf{G}$ and $\lambda_m^{\gamma+1/2} \mapsto c^{2\gamma+1}\lambda_m^{\gamma+1/2}$, canceling in numerator and denominator of (2) as $\epsilon_{\text{LT}} \to 0^+$. □

C. Anomaly Scoring and Operational Detection Latency

At test time, windows are scored at stride 1. Let $g_d : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ denote the dimension-agnostic coordinate saliency gate defined in (6). Unconstrained TS-JEPA scores the single-horizon residual $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t)$ via $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$. In Soliton-Reynolds JEPA, we combine the joint MD-dimensional whitened residual gate $g_{MD}(\tilde{\mathbf{e}}(t))$ with the peak sub-horizon gate $\max_{1\leq m\leq M} g_D(\tilde{\mathbf{e}}_m(t))$ into the multi-horizon envelope $\mathcal{S}_{\text{env}}(t)$, and modulate it by a one-sided median absolute deviation (MAD)-standardized soliton concentration gain:

$$\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}\Big(g_{MD}\big(\tilde{\mathbf{e}}(t)\big)^2 + \max_{1\leq m\leq M} g_D\big(\tilde{\mathbf{e}}_m(t)\big)^2 \Big)}, \tag{4}$$

$$\mathcal{S}_{\text{Sol-Rey}}(t) = \mathcal{S}_{\text{env}}(t)\Big(1 + \alpha_{\text{sol}} \max\big(0, \tanh(0.5\, z_{\text{LT}}(t))\big)\Big), \tag{5}$$

where $\alpha_{\text{sol}} = 0.12$ and $z_{\text{LT}}(t) = (R_{\text{LT}}(\mathbf{E}'(t)) - m_{\text{LT}})/(1.4826\,\text{MAD}_{\text{LT}})$ standardizes R_{LT} against its nominal training median m_{LT} and MAD MAD_{LT}. Because $\max(0, \tanh(0.5\, z_{\text{LT}}))$ is one-sided, nominal windows with below-median spectral concentration ($z_{\text{LT}} \leq 0$) receive zero amplification. Setting $\alpha_{\text{sol}} = 0$ recovers the unmodulated multi-horizon envelope control $\mathcal{S}_{\text{env}}(t)$ (Soliton-Reynolds (α_{sol}=0)).

Three-Part Operational Causality Framework: We formalize online evaluation under three criteria (Fig. 2):

(a) *Data causality ($\mathcal{F}_\tau$-measurability):* At decision timestamp $\tau = t + S - 1$, the detector observes only $\mathbf{X}[t - C : \tau, :]$; neither encoder accesses observations beyond τ.

(b) *Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):* Because z_{tgt} summarizes suspect observations $\mathbf{X}[t : t + S, :]$ spanning sample indices $[t, t + S - 1]$, score s_t is emitted at the trailing edge $\tau = t + S - 1$ (Fig. 2, Row 3), creating a structural emission lag of $S - 1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL, 63 min on GECCO, and 63 h on hourly Metro). This lag is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$ from onset t^* to the first alarm $t_{\text{alarm}} \geq t^*$. Shortening the horizon to $S \in \{1, 4\}$ eliminates emission latency for rapid tripwires (Section VII-D).

(c) *Strictly causal trailing vs. retrospective buffered attribution: Causal trailing attribution* assigns s_t only to $\tau = t+S-1$ for zero-lookahead streaming. *Retrospective buffered smear attribution* averages s_t across all suspect timestamps $[t, t+S-1]$, introducing fixed-lag smoothing. Both regimes are reported separately throughout (Table III(a),(b)).

D. Shared Architectural Components and Thresholding

1) Coordinate Saliency Gating and Scale Equivariance: To amplify sparse departures in high-dimensional representations while mitigating channel drowning, we define the coordinate saliency gate $g_d : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ for any vector $\mathbf{v} \in \mathbb{R}^d$:

$$g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}, \qquad w_i = \frac{\exp\big(|v_i|/(\tau_g \sigma_{\mathbf{v}})\big)}{\sum_{j=1}^{d}\exp\big(|v_j|/(\tau_g\sigma_{\mathbf{v}})\big)}, \quad \sigma_{\mathbf{v}} = \frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon, \tag{6}$$

with $\alpha = 0.5$, $\tau_g = 0.5$, and $\epsilon = 10^{-6}$. When $\mathbf{v} = \mathbf{0}$, $g_d(\mathbf{0}) = 0$. For non-zero $\mathbf{v}$, rescaling $\mathbf{v} \mapsto c\mathbf{v}$ with $c > 0$ yields $\sigma_{c\mathbf{v}} = c\frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon = c\sigma_{\mathbf{v}} + (1-c)\epsilon$. In the operational regime where $\|\mathbf{v}\|_2 \gg \epsilon$, the softmax weights w_i are scale-invariant up to an $\mathcal{O}(\epsilon/\|\mathbf{v}\|_2)$ perturbation, satisfying approximate degree-1 homogeneity $g_d(c\mathbf{v}) \approx c\, g_d(\mathbf{v})$ and recovering exact degree-1 homogeneity in the limit $\epsilon \to 0^+$. Input channels use uniform weights in the deployed pipeline (Section VI-B).

2) Multiscale Frequency Front-End: For multivariate streams ($K > 1$), a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Each projected channel is split across $J = 4$ smooth dyadic shells on a 256-point rFFT grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax scale router. Because smooth Fourier shells overlap at band edges ($\sum \Psi_j \equiv 1$ rather than $\sum \Psi_j^2 \equiv 1$), this acts as a multiscale front-end rather than an orthogonal Parseval wavelet bank (Table II(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ from nominal calibration data alone, SPOT [8] fits exceedances $Y = \mathcal{S}(t) - u > 0$ above the 98th calibration percentile $u = p_{98}$ to a Generalized Pareto Distribution (GPD) [38] with shape ξ and scale σ via Grimshaw maximum-likelihood estimation [49] with bounded L-BFGS-B fallback ($\xi \in [-0.9, 10]$). Inverting the GPD tail at target risk $q = 10^{-3}$ yields:

$$\tau_{\text{SPOT}} = \begin{cases} u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], & |\xi| \geq 10^{-5}, \\ u - \sigma \ln\left(\frac{qN_{\text{cal}}}{N_u}\right), & |\xi| < 10^{-5}, \end{cases} \tag{7}$$

where the lower branch is the continuous exponential-tail limit as $\xi \to 0$, evaluated directly to avoid division by zero. Here N_{cal} and N_u count post-attribution score timestamps on the calibration block. Primary tables use standard SPOT; Section VII-F compares parametric SPOT against non-parametric empirical quantiles, runs-declustered GPD fitting, and rolling thresholds.

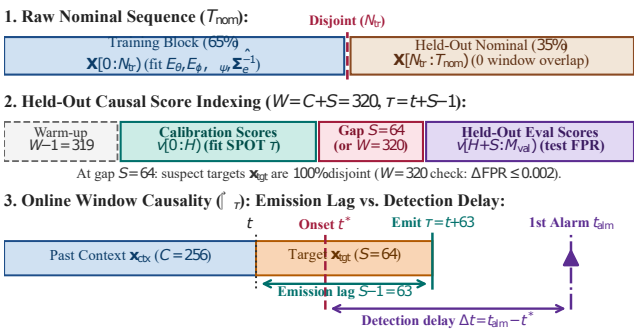


Fig. 2. Held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated calibration and evaluation blocks, and (Row 3) score-emission lag versus fault-detection delay.

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures use an identical streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$), trained at stride 16 and evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (64 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, standard deviation, Log-Ricci energy, and first-order total variation) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW [50] (learning rate $10^{-3} \to 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32), with identical training budgets and zero per-dataset hyperparameter tuning across families.

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W - 1 = 319$ observations provide context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid score timestamps $v[0 : M_{\text{val}})$. Splitting v at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $v_{\text{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\text{hold}} = v[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $v[H : H + S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint. The first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. A full-window isolation check ($G = W = 320$, excluding $v[H : H + 320)$ across the seven streams with $M_{\text{val}}/2 > 420$) confirms that shared passive

context does not bias calibration: stream-macro causal FPR changes by less than 0.25 percentage points for both JEPAs (0.33% to 0.36% for Soliton-Reynolds JEPA and 3.02% to 3.24% for TS-JEPA), remaining an order of magnitude below TimesNet and TranAD (9.7%–14.0%). All primary tables use fixed standard SPOT thresholds during testing.

b) Independent Statistical Units, Multi-Seed Replication, and Equivalence Bounds: Principal models on the 19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds ($\{7, 42, 2024\}$). To prevent pseudoreplication, seeds are averaged within each stream first (SD_{seed}), stream-level paired tests treat the telemetry stream ($n = 33$ primary or $n = 19$ non-GHL) as the unit of analysis, and domain-level tests treat the dataset family ($N = 7$) as the independent cluster:

1) *Paired Wilcoxon signed-rank test and false-discovery-rate adjustment:* The Wilcoxon signed-rank test evaluates whether the median paired difference across streams is zero without Gaussian assumptions. We adjust p-values across multiple comparisons via the Benjamini–Hochberg false-discovery-rate (FDR) procedure [51] ($q_{\text{FDR}} = 0.05$), report 95% stream-resampled bootstrap confidence intervals (CIs; 2,000 resamples), and cross-check at the dataset-cluster level ($N = 7$) via exact sign-flip permutation tests and zero-tuning LODO sensitivity.
2) *Two One-Sided Tests (TOST) for fixed-backbone regularizer equivalence:* We use TOST [15] to test whether the multiplicative soliton and manifold gain in (5) alters aggregate Point-F1 relative to the unmodulated multi-horizon whitened envelope $\mathcal{S}_{\text{env}}(t)$ on the identical trained backbone ($n = 19$ streams). TOST tests $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}), establishing equivalence when $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ across $\Delta \in \{\pm 0.015, \pm 0.020, \pm 0.030\}$ (Table II(a)).
3) *Distributional tail and drift diagnostics (γ_2 and D_{KS}):* On nominal windows we measure: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s - \mu)^4]/\sigma^4 - 3$, distinguishing heavy-tailed *leptokurtic* ($\gamma_2 > 0$) from light-tailed *platykurtic* ($\gamma_2 < 0$) residuals; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0, 1]$, quantifying distributional shift between calibration (F_{cal}) and held-out nominal (F_{hold}) blocks.

B. Baseline Fairness Checks, Cohort Structure, and Reproducibility

Two fairness audits verify that reconstructive baseline results under the unified 20-epoch, $W = 320$ pipeline are not artifacts of window length or implementation: (i) evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their native published window sizes across five streams raises Point-F1 to 0.164 and 0.218, but observation residuals remain severely heavy-tailed ($\gamma_2 > 98$ vs. 3.1 for TS-JEPA) with high nominal FPR (28.4%–44.1%); and (ii) reconstructors replicate their published point-adjusted F1 on SMD and MSL, while Table III(d) provides a secondary 130-stream, 11-dataset single-seed cross-check against NCAD, DCdetector, and Anomaly Transformer.

Experiments span five cohorts: *Cohort 1* (33 primary streams: 19 non-GHL at 5 seeds and 14 GHL at 3 seeds, Tables III(a)–(c) and V(a)); *Cohort 2* (19-stream, 12-arm matched-backbone factorial and operating-curve suite at 5 seeds, Table II(a),(c) and Fig. 4(b)); *Cohort 3* (6-stream single-channel quiet-fault and hybrid-monitor suite at 3 seeds, Fig. 3); *Cohort 4* (5-stream horizon-latency sweep across $S \in \{1, 4, 16, 32, 64\}$ and dyadic front-end ablation at 3 seeds, Fig. 4(a) and Table II(b)); and *Cohort 5* (141-stream, 3-seed SPOT benchmark across 12 multivariate datasets with validation early stopping plus the 130-stream single-seed baseline screen, Table III(d)). At $K = 25$ channels, online parameter counts are matched: 272k for TS-JEPA (540k with EMA target), 312k (619k total) for Soliton-Reynolds JEPA, 304k–307k for TCN observation forecasters, 649k for TimesNet, and 206k for TranAD. Non-finite entries are zero-imputed ($\boldsymbol{\sigma} \geq 10^{-2}$), and causal and buffered scores receive separately fitted thresholds τ_{SPOT} on v_{cal} (7). Runs executed on PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB memory); code is at https://github.com/quaid281/NCAD.

C. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [16], [40], we evaluate models across three tiers. First, *primary calibration and accuracy outcomes* comprise realized held-out nominal FPR (FPR_{hold}), out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ relative to design risk $q = 10^{-3}$, unadjusted Point-F1, and threshold-free PR-AUC. Second, *operating characteristics* evaluate event recall across deployable calibration quantiles (q^*, Fig. 4(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)). Third, *secondary operational endpoints* report unweighted Dataset-Macro Point-F1, event recall (fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\text{F1}_{\text{all-pos}} = 2\pi/(1 + \pi)$ at prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Multi-Horizon Whitened Envelope vs. Soliton Concentration Modulation

What actually drives the gains of Soliton-Reynolds JEPA over single-horizon TS-JEPA? To separate the core representation from post-hoc scoring multipliers on the exact same trained network, Table II(a) and Fig. 3(a) compare single-horizon TS-JEPA against the unmodulated multi-horizon envelope control (α_{sol}=0, which outputs $\mathcal{S}_{\text{env}}(t)$ directly) and a sweep across $\alpha_{\text{sol}} \in [0.03, 0.24]$ on all 19 non-GHL streams. Replacing single-horizon TS-JEPA with the unmodulated multi-horizon whitened envelope $\mathcal{S}_{\text{env}}(t)$ drives the core improvements in false-alarm control and low-FPR sensitivity (Table II(a),(c) and Fig. 3(a)): strictly causal held-out FPR drops four-fold from 2.4% to 0.6%, calibration drift decreases, and causal event recall at a 1% false-alarm budget rises from 0.395 to 0.578 without loss of buffered PR-AUC.

Across the modulation sweep $\alpha_{\text{sol}} \in [0, 0.24]$ (Fig. 3(a)), adding the post-hoc Lieb–Thirring soliton concentration gain

TABLE II
ARCHITECTURAL ABLATION AND MECHANISM-DISCRIMINATION STUDIES (n=19 STREAMS, 5 SEEDS; EVENT RECALL (ER)).

(a) Fixed-Backbone Soliton Control (n=19)

Model	Pt-F1	PR	FPR	ER	ΔF1 [90%]	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline	–
Soliton-Reynolds JEPA	0.148 ± 0.039	**0.331**	**0.078**	0.430	−0.001 [−.007, +.005]	$< 10^{-6*}$
$\alpha_{\text{sol}} = 0$ control	0.149 ± 0.038	0.330	0.083	0.462	–	$(< 10^{-4\dagger})$
TimesNet	0.197 ± 0.018	0.329	0.300	**0.583**	–	–
TranAD	0.130 ± 0.004	0.287	0.307	0.490	–	–

* TOST equivalence vs. α_{sol}=0 control at $\Delta = \pm 0.030$ ($p = 1.1 \times 10^{-7}$); † at $\Delta = \pm 0.020$ ($p = 2.7 \times 10^{-5}$; Wilcoxon p=0.472). Matched-FPR ER (f^*=0.01): causal 0.562 vs. 0.578; at f^*=0.05: 0.698 vs. 0.711.

(b) Dyadic Front-End (n=5)

Stream (F1 / PR)	Std. Conv	Dyadic	ΔF1 / ΔPR
SMAP P-3	**0.301** / 0.280	0.115 / **0.296**	−0.186 / +0.017
MSL M-1	0.014 / **0.491**	**0.016** / 0.449	+0.002 / −0.043
SMD machine-1-2	0.000 / 0.028	**0.068** / **0.221**	+0.068 / +0.193
Daphnet S01R01E1	**0.179** / **0.103**	0.130 / 0.066	−0.048 / −0.037
GECCO water-quality	**0.097** / 0.161	0.062 / **0.282**	−0.035 / +0.121
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	−0.040 / +0.050

(c) 12-Arm Matched-Backbone Factorial Comparison (n=19 Streams, 5 Seeds)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	**0.331**	0.487	0.730	0.395	0.584
Soliton-Reynolds JEPA	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	**0.331**	0.478	0.752	**0.562**	**0.698**
Soliton-Reynolds (α_{sol}=0)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	**0.578**	**0.711**
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Hard-Target Copy	1.16	0.167	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	0.816	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	**0.588**	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

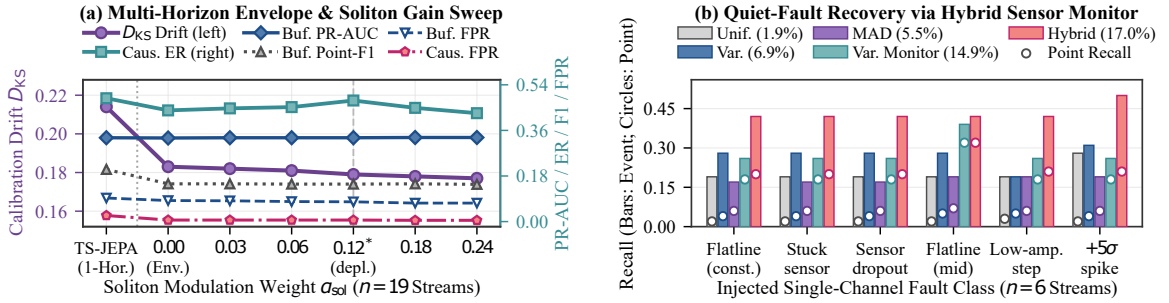


Fig. 3. Multi-horizon soliton modulation sweep and quiet-fault recovery. **(a)** Calibration-to-holdout drift (D_{KS}, left axis) and buffered PR-AUC, causal ER, buffered Point-F1, buffered FPR, and causal FPR (right axis) across single-horizon TS-JEPA and Soliton-Reynolds JEPA modulation weights $\alpha_{\text{sol}} \in [0, 0.24]$ (n=19 streams, five seeds). **(b)** Injected single-channel ER (bars) and point recall (circles) across uniform TS-JEPA (Unif.), context-variance (Var.) and median absolute deviation (MAD) input gates, the standalone rolling variance-collapse monitor (Var. Monitor), and the hybrid union detector (Hybrid; nominal FPR in parentheses, n=6 streams, three seeds).

on top of $\mathcal{S}_{\text{env}}(t)$ yields a modest false-alarm reduction and a null aggregate F1/PR-AUC response. At the deployed weight $\alpha_{\text{sol}} = 0.12$, buffered held-out FPR decreases slightly from 8.3% to 7.8% (Wilcoxon $p = 0.026$) and causal event recall rises from 0.439 to 0.478, while causal FPR and KS drift remain unchanged. Buffered Point-F1 and PR-AUC satisfy TOST equivalence relative to the unmodulated control within $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 2.7 \times 10^{-5}$, Table II(a)), and auxiliary spectral operator regularizers during training similarly leave downstream recall and FPR flat. The primary gains come from multi-depth TCN tapping, barycenter centering, and multi-horizon Mahalanobis whitening rather than post-hoc soliton modulation.

B. Ablation 2: Single-Channel Quiet Faults, Input Gating, and Hybrid Monitoring

Why do normalized latent encoders miss quiet single-channel faults? When one sensor out of K freezes at a nominal value while the remaining $K-1$ channels fluctuate normally, the frozen channel perturbs the pooled pre-LayerNorm vector by only $O(1/K)$. LayerNorm then rescales the embedding back onto the $\sqrt{D}$ sphere using the variance of the remaining $K-1$ active channels, masking the single-sensor freeze. Weighting input channels by context variance prior to encoding lowers macro Point-F1 across four multivariate benchmark streams from 0.118 to 0.077, so the deployed pipeline retains uniform input encoding. Across six fault classes injected into individual channels on six multivariate streams (Fig. 3(b)), uniform TS-JEPA detects 27.8% of abrupt $+5\sigma$ spikes, but on persistent quiet faults (constant or mid-range flatlines, stuck sensors, and zero dropouts) event recall falls to 19.4% and point recall drops below 3%, with input-channel gating offering little improvement. Simple observation-space monitors catch these complementary failures: a maximum channel deviation monitor (tracking the largest standardized deviation from the training median and MAD across channels) and a rolling variance-collapse monitor (tracking how far each channel's 32-step trailing standard deviation drops below its nominal median) each recover 25%–39% of quiet-fault events at roughly 14%–15% FPR. Taking the logical union of TS-JEPA alarms and the rolling variance-collapse monitor lifts event recall to 41.7%–50.0% across all six fault classes, trading a higher nominal FPR (17.0% vs. 1.9% for standalone TS-JEPA) for quiet-fault coverage.

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Table II(b) trains TS-JEPA with the identical 3-layer causal TCN encoder across five streams, comparing the 4-shell Fourier filterbank against a standard linear convolutional stem. Enabling the dyadic front-end raises macro PR-AUC from 0.213 to 0.263 (led by SMD and GECCO) and lowers validation residual kurtosis from 1.41 to 1.07, while macro Point-F1 decreases from 0.118 to 0.078 because the tighter tail produces a more conservative SPOT threshold on SMAP P-3. Frequency-shell splitting thus improves score ranking and tail conditioning rather than uniformly raising thresholded F1.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 12 model arms across all 19 non-GHL streams under identical chronological splits, training budgets, and matched parameter counts in Table II(c) isolates five architectural factors:

1) *Forecasting vs. reconstructing raw observations (TCN Obs. Forecaster vs. TCN Obs. Autoencoder):* Pairing the same causal dilated TCN encoder with a multilayer perceptron decoder to either forecast the 64-step target window or reconstruct the full 320-step window in raw sensor space leaves tail kurtosis ($\gamma_2 \approx 6.2$), calibration drift ($D_{\mathrm{KS}} \approx 0.45$), and causal FPR ($\approx 25\%$) nearly identical (Table II(c)). Switching from reconstruction to forecasting in observation space does not resolve threshold instability.
2) *LayerNorm bottleneck and saliency gating in observation space (TCN Obs. Forecaster (LN)):* Adding a $D = 32$ LayerNorm bottleneck, target-window LayerNorm, and the coordinate saliency gate to the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and cuts causal FPR in half to 11.7%. Because predictions still target raw sensor waveforms, however, calibration drift remains high ($D_{\mathrm{KS}} = 0.376$) and causal Point-F1 drops to 0.084.
3) *Learned latent target projection and multi-horizon whitening (TS-JEPA and Soliton-Reynolds JEPA):* Moving the prediction target from raw observations into the learned $D = 32$ latent space cuts calibration drift nearly in half relative to the bounded observation forecaster ($D_{\mathrm{KS}} = 0.214$ in TS-JEPA and 0.179 in Soliton-Reynolds JEPA, paired Wilcoxon $p \leq 0.001$). This lowers causal FPR to 2.4% and 0.6% while lifting buffered PR-AUC from 0.257 to 0.331.
4) *Nonlinear TCN encoding vs. linear PCA subspace and tree-based isolation (PCA Linear Forecaster (LN) and Isolation Forest):* Replacing the nonlinear TCN with 32 principal components and a ridge-regression predictor bounds causal FPR to 2.1% through LayerNorm and saliency gating, but causal Point-F1 collapses to 0.017 (and Isolation Forest reaches only 0.149 causal event recall). Learned nonlinear temporal features are therefore needed to separate anomalies from nominal background.
5) *Future prediction vs. context autoencoding, EMA target stability, and ℓ_2 scoring (JEPA Context Autoencoder, JEPA Hard-Target Copy, and TS-JEPA (ℓ_2 Scoring)):* Reconstructing the context window's own embedding in latent space keeps drift and causal false alarms low (1.7%), but drops causal Point-F1 from 0.161 to 0.112 and PR-AUC from 0.278 to 0.235, showing that forecasting future windows improves fault separation. Updating the target encoder by direct weight copying ($\zeta = 0$) or replacing the saliency gate with a Euclidean norm still holds causal FPR below 2% (0.9% and 1.8%), confirming that false-alarm reduction stems from the normalized latent representation rather than the scoring gate.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table III reports all 33 continuous telemetry streams under the chronological held-out SPOT protocol ($u = p_{98}, q = 10^{-3}$): Panel (a) uses retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns including anomaly event counts (E) and both causal and buffered held-out nominal FPRs. Evaluating realized FPR and out-of-sample calibration error $|\mathrm{FPR}_{\mathrm{hold}} - q|$ as primary outcomes shows a large relative reduction in false alarms and calibration error, while Soliton-Reynolds JEPA achieves the lowest held-out FPR across both attribution regimes (0.6% causal, 7.8% buffered), matches TS-JEPA on buffered PR-AUC (0.331), and leads in causal matched-FPR event recall (0.562 at $f^* = 0.01$).

Lower Realized FPR and Calibration Error on Held-Out Telemetry: On the 19 non-GHL streams (Table III(a),(b)), latent scoring cuts held-out nominal FPR and calibration error $|\mathrm{FPR}_{\mathrm{hold}} - q|$ three-fold under buffered attribution (7.8% for Soliton-Reynolds JEPA and 9.3% for TS-JEPA vs. $\approx 30\%$ for TimesNet and TranAD, $p \leq 0.045$). Under strictly causal streaming, the false-alarm reduction widens to ten- to forty-fold: 0.6% for Soliton-Reynolds JEPA and 2.4% for TS-JEPA, compared to 22.2%–24.3% for the reconstructors ($p = 0.010$). This separation holds across all 33 streams (0.38% and 1.51% causal FPR vs. 14.2%–16.1% for reconstructors, Table V(a)), across all $N = 7$ dataset families ($p = 0.016$), and under zero-tuning LODO evaluation (0.3%–2.3% causal FPR for JEPAs vs. 12.1%–18.4% for reconstructors). In Table III(c), Soliton-Reynolds JEPA achieves the lowest causal held-out FPR on every multi-stream family (0.1%–0.8% vs. 5.6%–36.0% for baselines).

Multi-Horizon Ranking Gains vs. Unadjusted Point-F1 Trade-offs: Under buffered attribution on the 19 non-GHL streams (Table III(a)), TS-JEPA and Soliton-Reynolds JEPA tie for the highest PR-AUC (0.331 vs. 0.329 for TimesNet and 0.287 for TranAD). Soliton-Reynolds JEPA trades lower buffered Point-F1 than TS-JEPA (0.148 vs. 0.206) for tighter false-alarm control, outperforming TranAD (0.130 F1) at one-quarter the FPR (7.8% vs. 30.7%). Across all 141 qualifying multivariate streams in 12 datasets at three seeds with validation early stopping (Table III(d)), 3-horizon Speculative-JEPA and Speculative-Alien-JEPA (adding a coreset and residual null-space bank) surpass single-horizon TS-JEPA both causally (0.184 vs. 0.165 PR-AUC, 95% CI $[+0.006, +0.034]$; 0.077 vs. 0.067 F1, with the Alien bank improving F1 on 74 of 122 non-tied streams, $p = 0.013$) and under buffered attribution

TABLE III
HELD-OUT CALIBRATION BENCHMARK UNDER SPOT ($u{=}p_{98}$, $q{=}10^{-3}$; BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution ($n{=}19$ Non-GHL)					(b) Causal Attribution ($n{=}19$ Non-GHL)				GHL ($n{=}14$)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093	**0.206**	**0.331**	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	**0.115**
Soliton-Reynolds JEPA	0.078	0.148	**0.331**	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Soliton-Reynolds ($\alpha_{\text{sol}}{=}0$)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive baselines												
TimesNet	0.300	0.197	0.329	**0.583**	**48.5**	0.243	**0.209**	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑				Held-Out FPR (Causal / Buffered) ↓			
Dataset	n	K	E	π	$\text{F1}_{\text{all-pos}}$	**TS-JEPA**	**Sol.-Rey.**	**TimesNet**	**TranAD**	**TS-JEPA**	**Sol.-Rey.**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.145	0.143	**0.229**	0.144	.030/.091	**.008/.072**	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	**0.022**	0.020	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.000	**0.012**	0.000	0.000	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.161**	0.081	0.158	0.151	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	**0.379**	0.267	0.289	0.167	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	**0.001**	**0.001**	0.000	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	**0.119**	0.008	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: 95% stream-bootstrap CIs ($n{=}19$) for (a) Buffered FPR/F1: TS-JEPA [.048, .145]/[.113, .319], Sol.-Rey. [.032, .137]/[.081, .229], ($\alpha_{\text{sol}}{=}0$) [.032, .145]/[.084, .226], TimesNet [.162, .446]/[.098, .315], TranAD [.177, .445]/[.053, .232], Linear AR [.001, .011]/[.012, .104]; (b) Causal FPR/F1: TS-JEPA [.007, .046]/[.063, .287], Sol.-Rey. [.003, .011]/[.013, .097], ($\alpha_{\text{sol}}{=}0$) [.003, .012]/[.014, .109], TimesNet [.123, .375]/[.109, .326], TranAD [.119, .337]/[.058, .241]. Causal alert rate $\hat{p}_{\text{alert}}$/TPR/$\text{ER}_{\text{rand}}(\text{FPR}_{\text{hold}})$ ($\text{ER}_{\text{rand}}(\hat{p}_{\text{alert}})$): TS-JEPA .108/.166/.283 (.656), Sol.-Rey. .024/.040/.272 (.491), ($\alpha_{\text{sol}}{=}0$) .025/.045/.278 (.487), TimesNet .242/.340/.600 (.801), TranAD .213/.239/.601 (.801). **(d) Archive Checks:** *130-Stream Screen (Pt-F1, macro/stream):* NCAD .058/.088; DCdetector .070/.097; Anomaly Trans. .091/.082; TimesNet .129/.151; TranAD .156/.152; TS-JEPA **.160/.186**. *141-Stream, 3-Seed SPOT (12 Datasets, Causal/Buffered PR-AUC [ER]):* TS-JEPA .165/.213 [.480/.459]; Speculative-JEPA .183/.**256** [.497/.565]; Speculative-Alien-JEPA .**184**/.253 [.**507**/.**567**].

(0.253–0.256 vs. 0.213 PR-AUC, $p \leq 0.0042$, and 0.567 vs. 0.459 event recall, $p = 1.6{\times}10^{-4}$), winning on 10 of 12 datasets led by SMAP ($n{=}47$, $0.308 \rightarrow 0.378$ buffered, $0.204 \rightarrow 0.252$ causal) and Exathlon ($n{=}17$, $0.175 \rightarrow 0.224$ buffered). On GHL ($\pi = 0.013$, Table III(c)), all JEPAs and the linear autoregressive (AR) forecaster fire near-silently (0.1%–0.4% FPR, 0.000 F1), showing that low FPR without recall does not indicate detection.

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.
2) *Latency-constrained telemetry:* Systems where a 63-step emission lag is prohibitive for immediate tripwire response (such as hourly Metro where 63 steps = 63 h), requiring short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Inactive sensor modes (flatlines, dropouts) that LayerNorm embeddings absorb as nominal background, requiring the hybrid union detector (combining TS-JEPA with a rolling variance-collapse monitor) when single-channel coverage is mandatory (Section VI-B).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Component-Level Analysis of Soliton-Reynolds JEPA

Tables II(a),(c) and III(a),(b) isolate how each component of Soliton-Reynolds JEPA modifies performance relative to TS-JEPA. First, *multi-depth TCN tapping and barycenter centering* remove common-mode baseline shift, lowering calibration drift (D_{KS} from 0.214 to 0.179) and cutting the inter-block drift share of excess causal FPR four-fold (71.3% to 18.7%, Table V(b)). Second, *nested multi-horizon rollout and Mahalanobis whitening* ($\mathcal{S}_{\text{env}}(t)$ at $\alpha_{\text{sol}} = 0$) capture sub-horizon transients across $S_m \in \{16, 32, 48, 64\}$, reducing causal FPR from 2.4% to 0.6% and lifting causal event recall at a 1% FPR budget from 0.395 to 0.578. Third, *post-hoc Lieb–Thirring soliton modulation* ($\alpha_{\text{sol}} = 0.12$) trims buffered FPR slightly (8.3% to 7.8%, $p = 0.026$) and nudges causal event recall from 0.439 to 0.478, but leaves Point-F1 and PR-AUC statistically unchanged on the fixed backbone ($p_{\text{TOST}} < 10^{-4}$), confirming that the multi-horizon whitened representation drives the primary gains.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Comparing Panels (a) and (b) of Table III shows how zero-lookahead streaming shifts the operating threshold. In causal mode, Soliton-Reynolds JEPA preserves competitive threshold-free ranking (PR-AUC 0.259 vs. 0.278 for TS-JEPA and 0.248 for TranAD). Without 64-step window averaging, however, causal 128-dimensional whitened residuals have a heavier right tail ($\gamma_2 = 2.30$) than buffered scores ($\gamma_2 = 1.25$) or single-horizon TS-JEPA ($\gamma_2 = 1.33$). When SPOT extrapolates that tail to $q = 10^{-3}$ in (7), it places τ_{SPOT} high on the causal score distribution. Consequently, Soliton-Reynolds JEPA alerts on only $\hat{p}_{\text{alert}} = 2.4\%$ of test steps (4.0% true-positive rate (TPR) at 0.6% held-out FPR), compared to 10.8% for TS-JEPA and 24.2% for TimesNet (Table III(b)). Because it flags only the sharpest peak steps inside each anomaly segment rather than staying on throughout the event, its causal Point-F1 drops

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION (5 SEEDS; EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**SPOT ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Soliton-Rey. JEPA	**0.652**	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Sol.-Rey. (α_{sol}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05	f^*=0.01	f^*=0.05	q^*=0.05	f^*=0.01	f^*=0.05
Soliton-Rey. JEPA	0.628	0.498	0.590	0.782	0.683	0.781
Sol.-Rey. (α_{sol}=0)	**0.640**	**0.540**	**0.603**	**0.785**	0.671	**0.785**
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 primary SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL streams into in-range contextual ($\max|z| \leq 5\sigma$) vs. out-of-range point ($\max|z| > 5\sigma$) causal events.

(0.050 vs. 0.161 for TS-JEPA) even though it catches 47.8% of anomaly events.

To check whether 47.8% event recall could arise by chance on long anomaly segments (median length $L = 139$ steps across 111 non-SWAN events), Table III (Note) compares each detector against an analytical rate-matched i.i.d. Bernoulli random alerter, $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$. At Soliton-Reynolds JEPA's held-out nominal rate ($p = 0.56\%$), random firing catches only 27.2% of events versus 47.8% actual; at its overall test firing rate ($\hat{p}_{\text{alert}} = 2.4\%$), unclustered Bernoulli trials reach 49.1% because real model alarms cluster inside detected events rather than firing independently. For TimesNet (24.3% held-out FPR), random firing alone reaches 60.0% event recall, exceeding TimesNet's actual 58.8%. Relaxing the static threshold to the empirical $p_{99.5}$ calibration quantile raises Soliton-Reynolds JEPA's causal event recall to 60.9% at 1.26% held-out FPR.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

Target horizon $S = 64$ imposes a 63-step emission lag before an alarm can trigger (Section IV-C). To evaluate detection quality versus latency against baselines at matched emission lag $S-1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero lag) and $S = 4$ (3-step lag), TS-JEPA keeps causal FPR at 0.4%–0.6% (vs. 2.5%–3.1% for TimesNet) while maintaining higher causal PR-AUC (0.178–0.189 vs. 0.150–0.162) and short detection delays ($\Delta t = 57.8$–85.2 steps). As S increases from 16 to 64, TimesNet causal FPR climbs from 6.1% to 21.3%, whereas latent predictors remain bounded ($\leq 2.0\%$). Consequently, $S = 64$ suits high-rate sensors (≥ 1 Hz, such as 64 Hz Daphnet where 63 steps = 0.98 s) or slow subsystem trend monitoring (0.1 Hz SMAP, MSL, and GHL, 10.5 min). Low-rate infrastructure streams ($\leq 1\,\text{min}^{-1}$, such as hourly Metro where 63 h lag is prohibitive) and rapid tripwires instead require $S \in \{1, 4\}$.

E. Operating Budgets and Contextual vs. Point Anomalies

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 0.3%–11% for latent predictors, and long segments (402 steps on Daphnet, 1,102 on SMAP) allow a 30% random-alarm detector to touch $> 95\%$ of segments via $1-(1-\text{FPR})^L$, we evaluate three comparisons:

1) *Deployable nominal calibration quantiles (q^*, Fig. 4(b), Table II(c)):* At causal calibration quantile $q^* = 0.050$ (set without test labels), TS-JEPA and Soliton-Reynolds JEPA achieve higher event recall (0.730 and 0.752) at lower false-alarm rates than TimesNet (0.618 recall) and TranAD (0.602 recall).
2) *Post-hoc matched realized test-FPR budgets (f^*, Fig. 4(b), Table II(c)):* Equalizing realized test nominal FPR across models (vertical dotted guides in Fig. 4(b)) proves that the JEPA detectors' low false-alarm rate is not an artifact of low alert frequency. At a 1% test FPR budget ($f^* = 0.01$), Soliton-Reynolds JEPA detects 56.2% of anomaly events (57.8% for α_{sol}=0) and TS-JEPA detects 39.5%, more than doubling TimesNet (21.7%) and TranAD (14.9%). A similar gap holds at $f^* = 0.05$ (69.8% vs. 34.3%–40.7%), whereas at relaxed budgets ($f^* \geq 0.10$) recall curves converge ($p = 0.625$).
3) *Contextual vs. point anomaly stratification (Table IV):* Out-of-range point anomalies (abrupt amplitude spikes exceeding $\pm 5\sigma$) can be flagged even by a memoryless marginal z-score (`ZScoreMax`), whereas contextual anomalies stay within normal sensor bounds ($\max|z| \leq 5\sigma$) and only violate multi-step trajectory structure. In Table IV(a), stratifying the 10 primary SMAP/MSL streams by NASA's official ground-truth taxonomy [4] shows that observation-space detectors collapse on contextual anomalies (0.161–0.317 buffered PR-AUC and 30%–56% recall at $f^* \in \{0.01, 0.05\}$). Soliton-Reynolds JEPA leads every contextual metric, achieving 0.652 buffered PR-AUC, 75.0% causal SPOT recall at 0.6% FPR, and 82.0%–92.0% recall at $f^* \in \{0.01, 0.05\}$ (surpassing TS-JEPA by 8–10 percentage points). Across all 19 non-GHL streams (Table IV(b)), multi-horizon Soliton-Reynolds JEPA and TS-JEPA nearly double the recall of reconstructors on in-range contextual faults ($\approx 60\%$ vs. 30% at $f^* = 0.05$) while also leading on point faults (78.1%–78.5% vs. 56.3%–61.1%).

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Static extreme-value thresholding does not achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample across all streams and attribution modes. Across all 33 streams (Table V(a)), standard SPOT achieves stream-macro causal FPR of 0.38% for Soliton-Reynolds JEPA and 1.51% for TS-JEPA, compared to 14.23%–16.11% for TranAD and TimesNet. Replacing parametric SPOT with a non-parametric empirical 99.5th percentile ($p_{99.5}$) yields nearly identical results (0.9%–2.3% causal FPR for JEPAs vs. 15.4%–17.5% for reconstructors), demonstrating that reconstructor false-alarm flooding is not an artifact of GPD tail fitting. Under $S = 64$ buffered attribution, however, smearing each window score

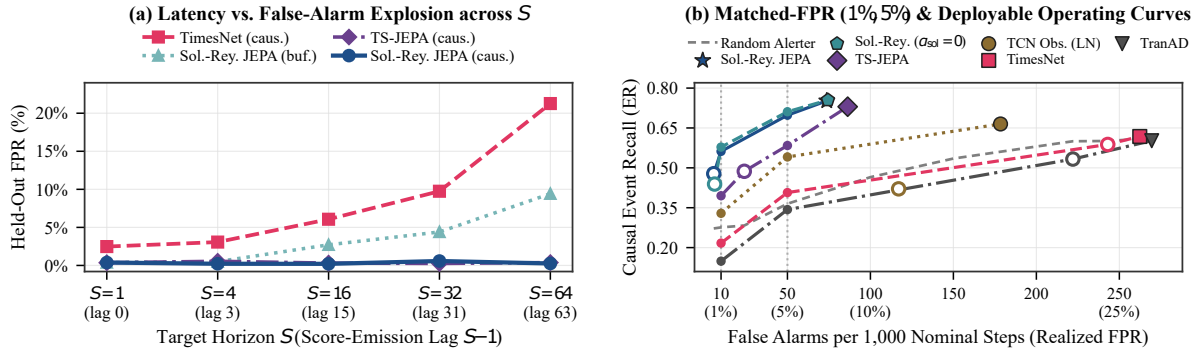


Fig. 4. Horizon length and operating trade-offs. **(a)** Held-out FPR across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched score-emission latency $S-1 \in \{0, 3, 15, 31, 63\}$ steps (five streams, three seeds). **(b)** Causal event recall versus false alarms across 19 non-GHL streams at equalized test-FPR budgets $f^* \in \{1\%, 5\%\}$ (vertical dotted guides) and deployable calibration quantiles from SPOT $q^*{=}10^{-3}$ (open circles) to $q^*{=}0.05$ (filled markers), plotted against a rate-matched Bernoulli random alerter.

TABLE V
THRESHOLD CALIBRATION AND DRIFT DECOMPOSITION ($q{=}10^{-3}$).

(a) Model	Rule	Causal FPR	Error (Over.)	Buf. FPR	Buf. F1
TS-JEPA	Standard SPOT	0.017 (0.015)	.014 (15×)	0.064 (0.055)	0.139
	Empirical p99.5	0.025 (0.023)	.022 (23×)	0.076 (0.066)	0.140
Sol.-Rey.	Standard SPOT	**0.004 (0.004)**	**.003 (4×)**	**0.055 (0.046)**	0.102
	Empirical p99.5	0.010 (0.009)	.008 (9×)	0.064 (0.054)	0.101
TimesNet	Standard SPOT	0.180 (0.161)	.160 (161×)	0.222 (0.199)	0.140
	Empirical p99.5	0.191 (0.175)	.174 (175×)	0.234 (0.213)	0.141
TranAD	Standard SPOT	0.160 (0.142)	.141 (142×)	0.222 (0.196)	0.094
	Empirical p99.5	0.171 (0.154)	.153 (154×)	0.234 (0.209)	0.096

	Refitted Causal FPR (n=19)			Excess Share (%)		
(b) Model	FPR_{chr}	FPR_{blk}	FPR_{pt}	Inter-Blk	Serial	i.i.d.
TS-JEPA	0.0229	0.0073	0.0033	71.3	18.1	10.5
Soliton-Rey.	**0.0056**	**0.0047**	**0.0020**	**18.7**	59.5	21.8
TimesNet	0.2302	0.0427	0.0035	81.8	17.1	1.1
TranAD	0.2103	0.0464	0.0034	78.3	20.5	1.1

In (a), cell-weighted means (n=33 streams; stream-macro in parentheses); Error is $|FPR_{macro} - q|$ and Over. is FPR_{macro}/q. In (b), 95% CIs for inter-block drift $FPR_{chr} - FPR_{blk}$: TS-JEPA $+.0156$ $[+.001, +.034]$, Sol.-Rey. $+.0009$ $[-.002, +.004]$, TimesNet $+.1874$ $[+.078, +.305]$, TranAD $+.1639$ $[+.072, +.267]$.

across 64 consecutive steps dilates every isolated exceedance into a 64-step cluster, inflating buffered FPR to 4.6%–5.5% for JEPAs and nearly 20% for reconstructors under both rules. Static SPOT is therefore near-calibrated for Soliton-Reynolds JEPA only in strictly causal mode. Two diagnostics explain why causal thresholds hold on latent residuals:

1) *Three-stage refitted permutation decomposition (Table V(b)):* To separate inter-block regime drift from serial correlation and GPD tail-fit error, we pool nominal calibration and evaluation scores $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}]$ across all 19 non-GHL streams and *refit* SPOT ($u = p_{98}, q = 10^{-3}$) under three splits: chronological (FPR_{chr}), $L = 64$ block permutation (FPR_{blk}, which removes macro-block drift while preserving intra-block serial correlation), and pointwise i.i.d. permutation (FPR_{pt}). Once temporal structure is removed by i.i.d. shuffling, every model achieves low false-alarm rates near the 0.1% target ($FPR_{pt} \leq 0.35\%$). In chronological deployment, inter-block drift ($FPR_{chr} - FPR_{blk}$) accounts for 71.3% of excess causal FPR in single-horizon TS-JEPA and 78.3%–81.8% in reconstructors. In barycenter-centered Soliton-Reynolds JEPA, by contrast, the inter-block drift share drops four-fold to 18.7% ($FPR_{chr} = 0.56\%$ vs. $FPR_{blk} = 0.47\%$, Table V(b)).
2) *GPD tail stability, declustering, and rolling adaptation:* On the 19 non-GHL streams at $u = p_{98}$, GPD fits converge with 0% solver fallback and low threshold uncertainty ($SE(\tau)/\tau \leq 3.5\%$). Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in only 4.4% of JEPA cells versus 34.4%–54.4% for reconstructors. Applying Ferro–Segers runs declustering ($r = 5$) [39] prior to GPD fitting trims stationary causal FPR to 0.20% for Soliton-Reynolds JEPA and 0.44% for TS-JEPA. Under discrete step, variance, and regime shifts across 16 streams, adaptive rolling quantiles hold causal TS-JEPA FPR below 0.40% (vs. 0.8%–3.0% for TimesNet), whereas continuous monotonic mean ramps defeat trailing rolling windows (> 32% FPR across all models).

VIII. CONCLUSION AND FUTURE WORK

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated extreme-value calibration across 33 continuous telemetry streams under a chronological held-out protocol:

1) *Factorial Mechanism Isolation and Multi-Horizon Whitening:* Matched-encoder ablations separate score bounding from representation learning: LayerNorm bottlenecks compress residual kurtosis (γ_2 from 6.10 to 0.91), learned latent projections cut calibration drift in half (D_{KS} from 0.376 to 0.214 in TS-JEPA), and multi-horizon barycenter whitening in Soliton-Reynolds JEPA lowers drift further to 0.179 at 0.6% causal FPR. On the fixed backbone, post-hoc Lieb–Thirring soliton modulation is statistically equivalent to the unmodulated multi-horizon envelope under TOST ($p_{\text{TOST}} < 10^{-4}$), and short horizons ($S \in \{1, 4\}$) eliminate the 63-step emission lag.
2) *Relative False-Alarm Control and Archive-Wide Generalization:* Latent scoring lowers held-out false alarms three- to four-fold under buffered attribution and ten- to forty-fold causally. Soliton-Reynolds JEPA matches TS-JEPA on buffered PR-AUC (0.331) and doubles reconstructor event recall at a 1% test FPR budget (56.2% vs. 21.7% for TimesNet), while a hybrid variance monitor recovers quiet single-channel faults (41.7%–50.0% event recall) and scaling to all 141 qualifying multivariate streams across 12 datasets confirms +11.7% causal and +18.9% buffered PR-AUC gains for multi-horizon Speculative-Alien-JEPA over single-horizon TS-JEPA.

3) *Calibration Drift and Adaptive Thresholding:* Static SPOT overshoots $q = 10^{-3}$ under buffered window smearing and by $15\times$ causally for single-horizon TS-JEPA, whereas barycenter-whitened Soliton-Reynolds JEPA reduces causal overshoot to 0.38% stream-macro FPR across 33 streams and cuts the inter-block drift share four-fold (71.3% to 18.7%). Runs-declustered SPOT and rolling quantiles narrow the remaining gap across step and variance shifts, motivating regime-conditioned conformal calibration.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[5] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[6] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[7] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[11] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[12] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.

[13] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.

[14] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[15] D. J. Schuirmann, "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.

[16] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[17] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[18] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[19] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[20] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[21] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[22] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[23] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[24] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[25] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[26] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[27] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[28] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[29] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[30] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[31] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[32] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[33] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[34] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[35] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[36] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[37] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[38] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[39] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[40] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[41] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[42] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[43] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[44] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[45] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[46] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[47] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.

[48] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[49] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.

[50] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[51] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.